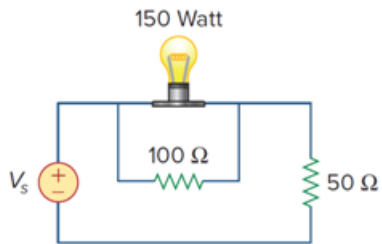


Problem

The 150 W light bulb in Fig. 2.122 is rated at 110 volts. Calculate the value of V_s to make the light bulb operate at its rated conditions.

Figure 2.122:



Step-by-step solution

Step 1 of 3

Consider the textbook Figure 2.122. The light bulb is 150 W and is rated at 110 V. The current through the bulb is,

$$\begin{aligned} I_{\text{bulb}} &= \frac{P_{\text{bulb}}}{V_{\text{bulb}}} \\ &= \frac{150 \text{ W}}{110 \text{ V}} \\ &= 1.364 \text{ A} \end{aligned}$$

Calculate the current through 100 Ω resistor.

$$\begin{aligned} I_{100 \Omega} &= \frac{V_{\text{bulb}}}{100} \\ &= \frac{110}{100} \\ &= 1.1 \text{ A} \end{aligned}$$

Step 2 of 3

Calculate the current through 50 Ω resistor.

$$\begin{aligned} I_{50 \Omega} &= I_{\text{bulb}} + I_{100 \Omega} \\ &= 1.364 + 1.1 \\ &= 2.464 \text{ A} \end{aligned}$$

Calculate the voltage across 50 Ω resistor.

$$\begin{aligned} V_{50 \Omega} &= (2.464)(50) \\ &= 123.2 \text{ V} \end{aligned}$$

Step 3 of 3

The expression for source voltage is,

$$\begin{aligned} V_s &= 110 + 123.2 \\ &= 233.2 \text{ V} \end{aligned}$$

Therefore, the voltage V_s to make the light bulb operate at its rated conditions is 233.2 V.

